Inflation, Interest Rates and the Predictability of Stock Returns ⋆

Martin Časta[1]

[a]*Czech National Bank, Na Prikope 28, Prague, 11000, Czech Republic*
[b]*Prague University of Economics and Business, nam. Winstona Churchilla, Prague, 12000, Czech Republic*

Abstract

This paper focuses on the predictive role of inflation and interest rates in forecasting stock returns. We report economically significant in-sample evidence that stock returns are predictable using cyclical components of inflation and interest rates. The out-of-sample analysis results show that these cyclical components have reasonable out-of-sample predictability for the equity risk premium when the comparison is made against a historical average benchmark. We believe this is due to the fact that both inflation and interest rates contain stochastic trends. Accounting for changes in trend inflation (and equilibrium real interest rate) is essential for forecasting stock returns.

Keywords: Cyclical Components of Interest Rates and Inflation, Predictability of Excess Stock Returns
JEL: G10, G12, G17

1. Introduction

Forecasting stock returns has long been recognized as a highly challenging task, and achieving consistent out-of-sample out-performance against average historical return benchmarks remains highly difficult. This well-known fact

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Email address: `martin.casta@cnb.cz` (Martin Časta)

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has not changed much since the seminal works of Goyal and Welch (Goyal and Welch, 2003, 2008; Goyal et al., 2021). Conversely, numerous studies assert the existence of predictive power in certain variables. These studies report economically significant in-sample evidence that stock returns are predictable. However, the out-of-sample performance is often mixed. But even in this case, the results differ significantly depending on the methodology and variables used, as is shown for example in Rapach et al. (2010), Dangl and Halling (2012), Pettenuzzo et al. (2014), Yin (2019), Pan et al. (2020), Tsiakas et al. (2020) and Wang et al. (2021). Overall, the predictability of stock returns has been the subject of much research and debate.

A wide range of economic variables has been proposed in the literature as potential predictors[2] of stock returns. In this article, our primary focus is on nominal interest rate variables and the inflation rate. Previous research by Campbell (1987), Fama and French (1989), Keim and Stambaugh (1986), Ang and Bekaert (2007), among others, have shown that stock returns can be predicted using nominal interest rates. Similarly, Nelson (1976), Fama and Schwert (1977), Campbell and Vuolteenaho (2004), and others have identified predictability based on the inflation rate. However, the forecasting power of these variables has deteriorated significantly since the 1980s (Goyal et al., 2021).

This article aims to explain the diminishing forecasting power of interest rates and inflation. In line with Denga et al. (2022), we emphasize the significance of expectations in asset pricing, as investors price assets based on their beliefs. In more detail, the agents distinguish between long-run trends and cyclical (transitory) fluctuations. According to asset pricing theory (Rapach and Zhou, 2013), the predictability of stock returns can arise from the influence of time-varying aggregate risk. Therefore, if forecasting models consistently incorporate this time-varying aggregate risk premium, their effectiveness will likely endure over extended periods. This reinforces the idea that incorporating the influence of such risk factors is crucial for maintaining forecasting model effectiveness in predicting stock returns.

Our study builds upon this premise, asserting that cyclical deviations from the long-term trends of standard macro-predictors significantly influ-

[2]Note that this list is not comprehensive, and many other predictors were suggested. For more information, see Rapach and Zhou (2013), where authors provide an extensive survey of the literature.

ence investors' risk perception. This is because the trends regarding inflation and interest rates consist of (slowly changing) steady-state expectations of investors, which are reflected in the valuation of all instruments on the financial market. On the other hand, short-term deviations from these trends primarily affect the risk aversion of individual agents, for example, because the cyclical fluctuations represent expectation forecast errors[3], or due to the existence adjustment cost, etc. For a theoretical explanation of cyclical variation in risk aversion and prices of risky assets, see, for example, Campbell and Cochrane (1999) or Campbell et al. (2017). Alternatively, we can simply state that cyclical components of interest rates and inflation serve as proxies for changing risk aversion.[4]

More specifically, inflation has, in theory, no effect if it affects nominal dividend growth rates and the nominal discount factor in the same way[5]. However, as our article demonstrates, the cyclical component of inflation negatively affects excess returns, indicating that these cyclical fluctuations influence risk aversion and enable the prediction of excess returns. On the other hand, it is widely acknowledged the stock can be considered an inflation hedge in the long run (Boudoukh and Richardson, 1993). This is possible only if the nominal dividend growth rate slowly adjusts to the inflation trend, which is reflected in changing nominal stock returns and stable real returns. In other words, we state that inflation can be described as an I(1) series and its long-term trend also affects other variables, i.e., interest rates load upon the inflation trend. Accounting for changes in trend inflation is, therefore,

[3]For instance, because agents update their information set sluggishly or due to how the expectations are formed, i.e., extrapolative/adaptive expectations, habit formation, behavioral inattention, etc.

[4]From an empirical point of view, the models may suffer from endogeneity issues as changing risk aversion may affect both dependent and explanatory variables. Also, given the fact that these cyclical components may serve as proxies for some other unobserved factor, the result may not indicate true causal relation. For example, Baker et al. (2019) and Chiang (2023) argue that inflation tends to co-move with stock market volatility, which has a negative effect on stock returns.

[5]In contrast, inflation has a negative impact on bonds, whose payoffs are set in nominal terms. An ambiguous impact on stocks is caused by the fact that their payoffs vary in relation to changes in inflation. However, inflation also influences the nominal pricing kernel, and the impact of inflation, therefore, depends on the relative sizes of these effects. Our preferred explanation of the results obtained below is that by focusing on cyclical components, we obtain variables that mainly influence the pricing kernel and cause changes in time-varying risk aversion.

essential for understanding stock returns.

A potential criticism of our approach is that the FED following inflation target suggests that inflation is stationary. However, this was not always the case; and the inflation targeting itself depends on the FED's credibility. Our reasoning also aligns with the changing stock-bond correlation discussion (Andersson et al., 2008; Pericoli, 2020). The negative correlation between stock and bond returns seen in the last two decades could be caused by the stabilization of long-term inflation expectations. Therefore, results concerning last two decades are mainly driven by cyclical fluctuations. Conversely, the positive correlation witnessed in the latter half of the twentieth century combined cyclical movements with simultaneous shifts in long-term expectation, making its interpretation highly complex. For example, Andersson et al. (2008) argues that stock and bond prices move in the same direction during periods of high inflation expectations, while epochs of negative stock-bond return correlation seem to coincide with subdued inflation expectations.

Differing from Denga et al. (2022), our study does not rely on survey data for extracting expectations but instead employs an exponentially weighted moving average (EWMA) detrending method to decompose investor expectations (regarding inflation and interest rates) into long-term[6] and cyclical components. This can be justified by assuming, for example, extrapolative expectations of inflation and interest rates.[7]

In this regard, our work draws inspiration from the shifting endpoint literature, exemplified by studies conducted by Kozicki and Tinsley (2001), Cieslak and Povala (2015), Bauer and Rudebusch (2020). These studies illustrate that empirical properties, interpretability, and predictions significantly improve when shifting endpoints are included, i.e., detrending. For instance, Kozicki and Tinsley (2001) demonstrate that a substantial portion of inflation rate variation can be explained by the inflation trend. Similar findings are observed for bond yields in the study by Cieslak and Povala (2015). Bauer and Rudebusch (2020) argue for including the change in real interest rate trend to fully capture the trend component in interest rates, as accounting solely for the inflation trend leaves a highly persistent component of interest rates unexplained. Our study also shares similarities with the

[6]This approach allows for steady states to be time-varying.

[7]Barsky and De Long (1993) and Evans and Honkapohja (2009) show that, in many cases, this approach coincides with rational expectations and represents the optimal prediction rule.

works of Cooper and Priestley (2009), Yu et al. (2023) and Atanasov et al. (2020), where authors also concentrate on the cyclical components of commonly used explanatory variables and highlight their enhanced predictive power compared to non-detrended variables.

The remainder of this article focuses on an in-sample and out-of-sample forecasting exercise for excess returns. We employ a standard bivariate regression approach, following the methodology outlined by Goyal and Welch (2008).

2. Data

We use the well-known dataset of Goyal and Welch (2008). More specifically, we work with updated (monthly) data for the variables used in their study, which are available from Amit Goyal's webpage. This is to establish a connection between our discoveries and the extensive literature studying the predictability of market returns. In other words, this allows us to conduct a comparative analysis between the forecasting capability of our predictors and other predictor variables outlined by Goyal and Welch (2008). More specifically, we incorporate the following predictors: the interest rate on a three-month Treasury bill (secondary market) (TBY) and long-term government bond yield (LTY). To measure the inflation rate, we use the year-on-year inflation rate calculated from the Consumer Price Index (π) for all urban consumers obtained from the U.S. Bureau of Labor Statistics. This is because year-over-year change is the preferred inflation measure closely watched by markets and central bankers. For comparison, we also use the inflation rate (π^{GW}) from Goyal's dataset. Consistent with the existing literature, we focus on predicting the log excess return on an S&P 500 Index. We compute excess returns by subtracting the return on the Treasury bill (a proxy for risk-free rate) from the market return.

As explained above, we assume that interest rates and inflation variables can be described as I(1) series.[8] In more detail, we primarily assume that both inflation and interest rates load upon the inflation trend. We follow Cieslak and Povala (2015) approach to obtain the inflation trend. Specifically, we construct the inflation trend as exponentially-weighted moving average of

[8]This is generally confirmed by ADF and KPSS test results.

past values of year-over-year inflation:

$$\pi_t^{trend} = (1-\lambda)\sum_{i=0}^{t-1}\lambda^i \pi_{t-i} \tag{1}$$

where π_t denotes year-over-year inflation, π_t^{trend} is inflation trend. We set λ to 0.987 as in Cieslak and Povala (2015). The detrended variables are obtained by subtracting the inflation trend. This can be written as follows: $TBY_t^{detr} = TBY_t - \pi_{t-1}^{trend}$, $LTY_t^{detr} = LTY_t - \pi_{t-1}^{trend}$, $\pi_t^{detr} = \pi_t - \pi_t^{trend}$.[9]

Furthermore, following Bauer and Rudebusch (2020), we also account for the shift in real natural/equilibrium interest rate. The change in real interest rate trend is again computed using EWMA:

$$r_t^{trend} = (1-\lambda)\sum_{i=0}^{t-1}\lambda^i r_{t-i} \tag{2}$$

where $r_t = TBY_t - \pi_t$ and we set λ to 0.999. This alternative detrending is obtained as follows: $TBY_t^{detr2} = TBY_t - r_{t-1}^{trend} - \pi_{t-1}^{trend}$, $LTY_t^{detr2} = LTY_t - r_{t-1}^{trend} - \pi_{t-1}^{trend}$.

The analysis is performed on the monthly[10] data ranging from 1951 (M1) to 2022 (M12). This is due to the fact that interest rate data are hard to interpret before the 1951 Treasury Accord, as the FED effectively pegged interest rates during the 1930s and the 1940s. We, therefore, focus on the post-Accord period.

3. Methodology

We consider following standard univariate predictive regression model to analyze aggregate excess stock return predictability:

$$r_{t+h}^e = \alpha + \beta x_t + \epsilon_{t+h} \tag{3}$$

where $r_{t+h}^e = r_{t+h} - r_{f,t+h}$ denote the equity risk premium at time $t+h$, r_{t+h} is the total log return (including dividends) at time $t+h$, $r_{f,t+h}$ is the

[9] We account for the delay in CPI data releases when testing the predictive ability of inflation and lagged this variable by an additional month.

[10] For forecasting excess returns above a one-month period, we use overlapping observations.

log return of risk-free rate, i.e., we measure r^e_{t+h} as the h-period continuously compounded log return less the corresponding h-period continuously compounded log risk-free rate. x_t is a predictor at time t, and ϵ_{t+h} denote error term. The coefficients α and β are estimated with OLS.

Due to well-known biases in standard OLS regressions, given the use of persistent explanatory variables, we use Newey-West standard errors. Furthermore, we calculate wild bootstrap p-values with the null hypothesis of unpredictable returns. This is because Newey-West standard errors deteriorate with longer horizon forecasts, especially when overlapping observations are used. The procedure account for persistence in explanatory variables and correlations between excess return and explanatory variables. The bootstrap procedure we apply follows that of Maio (2013).

As is standard in stock return literature, we run an out-of-sample forecasting exercise using a recursive window estimation, i.e. for every period, we add the actual realization of the excess return and other explanatory variables (interest rate, inflation) as additional data points at the end of the sample, and re-estimate the model. An h period-ahead forecast is generated at each step. We use the sample of the first 72 periods (6 years) as a training period for the estimation.[11] Following Goyal and Welch (2008), the historical average of the excess returns is used as a benchmark forecast model. The average is also estimated using a recursive window (since 1927).

Following Campbell and Thompson (2008); we also introduce an economically-motivated restriction on individual predictive regression models, which restricts the sign of the predictor coefficients and forecasted excess returns. According to this approach, the predictor coefficients must have the theoretically expected sign, and the equity risk premium must be positive. Placing the above-mentioned restrictions may somewhat improve out-of-sample performance.

To compare the out-of-sample forecasts, we use out-of-sample R^2 statistic (R^2_{OS}). The R^2_{OS} statistic has a similar properties to the standard in-sample R^2 statistic and can be written in following way:

$$R^2_{OS} = 1 - \frac{MSFE_{model}}{MSFE_{benchmark}} \tag{4}$$

[11] The length of the training period balances having enough observations for a reasonably precise estimation of the initial parameters with a relatively long out-of-sample period for forecast evaluation. Most of the studies set an initial period in the range of 5 to 15 years.

where $MSFE_{model}$ and $MSFE_{benchmark}$ denotes the mean-square-forecast error of the predictive regression and the benchmark historical average model. When $R^2_{OS} > 0$, the predictive regression model have a lower $MSFE$ than the benchmark.

The test statistics[12] (CW) by Clark and West (2007) was also employed for a more formal approach. The null hypothesis is that the root-mean-square-forecast error (RMSFE) of the predictive regression model and the historical benchmark are equal. The alternative hypothesis is that the predictive regression outperforms the historical benchmark.

4. Results

Table 1 display the in-sample predictive regression results. The table is divided into two panels for one-month (Panel A) and three-month (Panel B) excess returns[13]. The results regarding different predictors are shown in separate columns. In every column, we show the coefficient of the point estimate, the corresponding Newey-West standard error (in parentheses), bootstrapped p-value of the regression coefficient (square brackets), and the adjusted and un-adjusted R^2.

We find that the estimated coefficients regarding the transitory components of interest rates and inflation are negative and have an economically sizable impact on future excess returns. The coefficients' estimates are strongly statistically significant in both one-month and three-month horizons, and the associated R^2 is significantly higher than that of the non-detrended predictors.[14] The results are also confirmed by the bootstrapped p-values of the regression coefficients. Table 1 (Panel B) also shows that predictability is increasing with the horizon both in terms of the size of the estimated coefficients and R^2 statistics. Thus, expected returns are high when cyclical components of inflation and interest rates are low. Therefore, the periods of cyclical low inflation and cyclical low-interest rates (relative to the perceived long-term trends) are also periods of high-risk aversion and slower/negative (more uncertainty about) economic growth, i.e. high excess returns are gen-

[12] We used the HAC estimator to calculate the variance.

[13] Due to the well-known problems with long-horizon forecasts, we do not include them. See Boudoukh et al. (2008)

[14] Accounting also for the change in real interest rate trend does not bring significant benefits relative to including only inflation trend.

erally associated with a negative output gap, lower inflation and low nominal interest rates. In a similar fashion, Andrle et al. (2017) show that regarding business cycle fluctuations, there is mostly positive co-movement of output, interest rate and inflation. This can be explained by the fact that the main driver of business cycle fluctuations are demand shocks. Pericoli (2020) similarly argues that the dynamics of inflation has gone from counter-cyclical to pro-cyclical and asses that inflation is likely to be pro-cyclical when it is propelled by demand rather than supply shocks.

Table 1: In-Sample Results

	Panel A: One-Month								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
π^{detr}	-0.198^{**}								
	(0.077)								
	[0.0087]								
π		-0.137^{**}							
		(0.061)							
		[0.0283]							
π^{GW}			-0.920^{**}						
			(0.462)						
			[0.0322]						
TBY^{detr}				-0.232^{***}					
				(0.070)					
				[0.0155]					
TBY^{detr2}					-0.243^{***}				
					(0.069)				
					[0.0123]				
TBY						-0.116^{**}			
						(0.049)			
						[0.0702]			
LTY^{detr}							-0.260^{***}		
							(0.094)		
							[0.0286]		
LTY^{detr2}								-0.245^{***}	
								(0.086)	
								[0.0529]	
LTY									-0.090^{*}
									(0.054)
									[0.1636]
$Constant$	0.005^{***}	0.010^{***}	0.008^{***}	0.006^{***}	-0.001	0.010^{***}	0.011^{***}	0.003^{**}	0.011^{***}
	(0.001)	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)
Observations	863	863	863	863	863	863	863	863	863
R^2	0.012	0.008	0.006	0.013	0.014	0.007	0.010	0.009	0.004
Adjusted R^2	0.010	0.007	0.005	0.012	0.013	0.006	0.009	0.008	0.002

	Panel B: Three-Month								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
π^{detr}	-0.551^{**}								
	(0.214)								
	[0.0002]								
π		-0.378^{**}							
		(0.166)							
		[0.0023]							
π^{GW}			-2.391^{*}						
			(1.316)						
			[0.0101]						
TBY^{detr}				-0.611^{***}					
				(0.179)					
				[0.0006]					
TBY^{detr2}					-0.640^{***}				
					(0.175)				
					[0.001]				
TBY						-0.304^{**}			
						(0.126)			
						[0.0094]			
LTY^{detr}							-0.694^{***}		
							(0.228)		
							[0.0017]		
LTY^{detr2}								-0.651^{***}	
								(0.212)	
								[0.0049]	
LTY									-0.239^{*}
									(0.136)
									[0.0434]
$Constant$	0.016^{***}	0.030^{***}	0.023^{***}	0.019^{***}	0.001	0.029^{***}	0.031^{***}	0.011^{**}	0.030^{***}
	(0.004)	(0.006)	(0.006)	(0.004)	(0.006)	(0.006)	(0.006)	(0.005)	(0.008)
Observations	861	861	861	861	861	861	861	861	861
R^2	0.029	0.021	0.014	0.031	0.032	0.016	0.023	0.021	0.008
Adjusted R^2	0.028	0.020	0.012	0.030	0.031	0.015	0.022	0.020	0.007

Note: The table presents one-month ahead (Panel A) and three-month ahead (Panel B) predictive regressions for the equity risk premia. The Newey-West standard errors (in parentheses) with h lags were used. π is inflation rate computed from Consumer Price Index, TBY denotes the three-month treasury bill, and LTY is the long-term government bond yield. The analysis is performed on the data ranging from 1951 (M1) to 2022 (M12). *p<0.1; **p<0.05; ***p<0.01 (Newey-West standard errors). Bootstrapped p-values of the regression coefficients are displayed in square brackets.

Table 2 shows the results of the out-of-sample exercise. The table is again divided into two panels for one-month and three-month forecast horizons. In every panel, the first two rows display out-of-sample R^2_{OS} and the p-value of Clark West (CW) test statistics. The third and fourth rows show the same statistics but take into account Campbell and Thompson (2008) restrictions. (We set the regression coefficient to zero whenever it has the "wrong" sign and rule out a negative risk premium.) This out-of-sample forecasting exercise shows that the detrended interest rates and inflation have reasonable out-of-sample predictability for the risk premium compared to the historical average benchmark and the non-detrended predictors. This is true even for the one-month forecast horizon. Campbell and Thompson (2008) argue that even small positive R^2_{OS}, such as 0.5% regarding monthly data, signal an economically meaningful degree of return predictability. In our case, the detrended treasury bill yield satisfies this condition even without taking into account Campbell and Thompson (2008) restrictions.

Table 2: Out-of-Sample Results

	Panel A: One-Month								
	π^{detr}	π	π^{GW}	TBY^{detr}	TBY^{detr2}	TBY	LTY^{detr}	LTY^{detr2}	LTY
R^2_{OS}	0.004	0.000	-0.002	0.005	0.006	-0.010	0.002	0.002	-0.012
CW	0.103	0.199	0.106	0.007	0.012	0.017	0.074	0.120	0.167
$R^2_{OS,Campbell-Thompson}$	0.007	0.006	-0.001	0.004	0.005	0.004	0.003	0.003	0.001
$CW_{Campbell-Thompson}$	0.012	0.020	0.110	0.014	0.018	0.020	0.075	0.109	0.107
	Panel B: Three-Month								
	π^{detr}	π	π^{GW}	TBY^{detr}	TBY^{detr2}	TBY	LTY^{detr}	LTY^{detr2}	LTY
R^2_{OS}	0.020	0.011	0.002	0.018	0.020	-0.011	0.014	0.013	-0.010
CW	0.023	0.047	0.031	0.001	0.001	0.001	0.007	0.007	0.011
$R^2_{OS,Campbell-Thompson}$	0.025	0.020	0.003	0.012	0.015	0.015	0.016	0.016	0.012
$CW_{Campbell-Thompson}$	0.020	0.035	0.067	0.011	0.009	0.013	0.019	0.015	0.020

Note: The table shows the results of the out-of-sample forecasting exercise regarding one-month ahead (Panel A) and three-month ahead (Panel B) equity risk premia. π is inflation rate computed from Consumer Price Index, TBY denotes the three-month treasury bill, and LTY is the long-term government bond yield. CW denote the p-value of Clark West test statistics. Subscript Campbell-Thompson denote that restrictions as in Campbell and Thompson (2008) were used. The analysis is performed on the data ranging from 1957 (M1) to 2022 (M12).

5. Conclusion

Goyal and Welch (2008) showed that standard macro-finance predictors mostly have poor out-of-sample properties. This article attempts to show that this does not always have to be the case. More specifically, this paper's main contribution to the existing literature is an in-sample and out-of-sample forecasting exercise using a standard bivariate regression approach following Goyal and Welch (2008), where we illustrate that there exists substantial in-sample and out-of-sample predictability for the equity risk premium using cyclical components of inflation and interest rates when the comparison is made against a historical average benchmark. We also observe a significant forecast improvement in contrast to the non-detrended variables. The results can be rationalized by the existence of time-varying risk premia and the fact that inflation and interest rates contain stochastic trends.

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